

Student's guide

CESAR Science Case - The differential rotation of the Sun and its Chromosphere

Name

Date

Introduction

The Sun as you may already know, is not a solid body. It is a massive body of gas constantly active. Even if we cannot see it, it is concluded by observations that the Sun rotates at different velocities depending on where you are looking at. The poles actually rotate slower than the equator, which is not obvious to our eyes but measurements and observations have concluded that.

There is also many different ways to observe the Sun. Some of its layers can only be observed from Earth using different types of telescopes such as the h-Alpha telescope that allows us to look at the chromosphere, the layer of the Sun where protuberances are visible. Other telescopes just reduce the incoming light of the Sun allowing us to observe the photosphere that is the Sun's surface.

In the photosphere there are a huge amount of sunspots while a solar maximum occur. These sunspots are incredible big compared with human distances such as: a car, a house, a country.

In this laboratory, you are going to observe the Sun by taking images of it with different telescopes. The visual telescope will allow you to see how the Sun rotates at different velocities depending on the latitude. Using a h-Alpha telescope you are going to analyse the chromosphere trying to visualize its main structures such as: the protuberances, the filaments or the spicules.

Remember that if you want to observe the Sun by your own, you always have to use an appropriate filter, or project it on a screen.

Theoretical background

It was mentioned in the introduction that the Sun rotates at different velocities. The reason is because the Sun is not a solid body so the plasma and gases that are in constantly movement are not attached to anything and they are free to move anywhere. Convection movements below the surface and temperature differences push these masses to move faster in the equator and slower in the poles.

This particular rotation has very important consequences on the stars. For example it causes disturbance in its magnetic fields, causing the lines to "tangle". This can be read in the CESAR Booklet for this Science Case, but here is a part from the booklet related to this topic:

It takes about 25 days around the equator compared to 35 days at higher altitudes. With the equator moving in a faster pace than the other areas of the Sun, the magnetic field lines will twist together over time. This will then produce loops in the magnetic field, which in turn will erupt from the surface of the Sun and create the formation of the Sun's intense sunspots and other known solar events.

The easiest way to measure the Sun's differential rotation is by tracking its sunspots. Sunspots at different latitudes will rotate at different velocities. But, what is exactly a sunspot? It was introduced in the booklet for this laboratory, but it is worth mentioning it here as well:

A sunspot is a region of the Sun that is disturbed by powerful magnetic fields. These magnetic fields avoid the natural movements of the material cooling the area. As the temperature decrease, it gets darker. Typically not only one sunspot is generated in the same area. The sunspots appear in groups of two, five or even more than twenty.

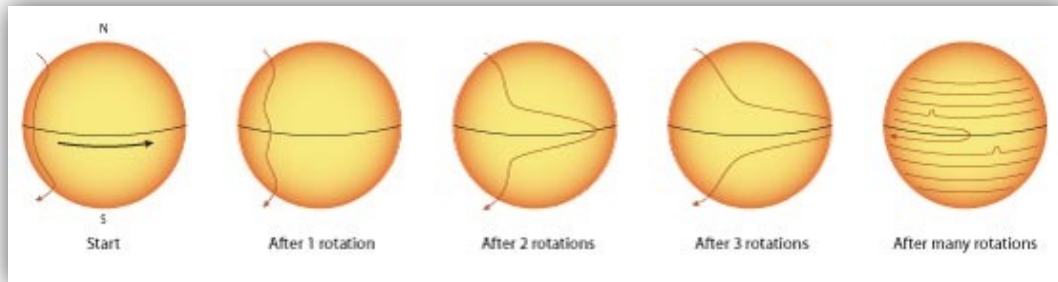
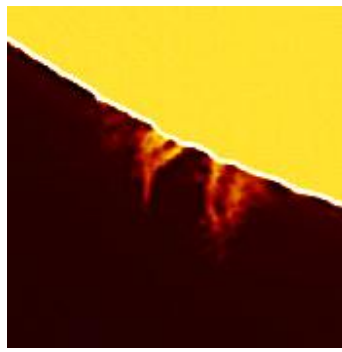


Figure 1: The differential rotation of the Sun and the creation of sunspots

Credit: NASA/IBEX

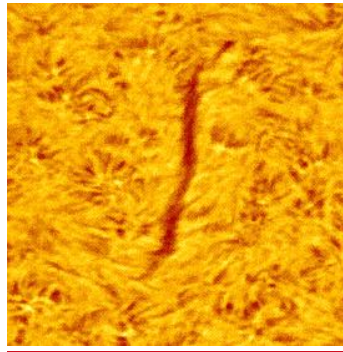
The sunspots can also be observed with an h-Alpha telescope but these telescopes are prepared to observe another details of the Sun, those that occur in the chromosphere. The chromosphere is a thicker layer than the photosphere and there is a huge amount of characteristics on it. To distinguish every structure of this layer, here is a short description for some of them:

- **Prominences:** they are one of the biggest structures from the Sun's surface. They usually appear with a loop shape around the solar disk. That shape is produced by the magnetic lines that go outside the Sun's surface. The material follows those lines so you could see the loop. Sometimes the loop is broken so what you see is just a mass ejection without any particular shape.



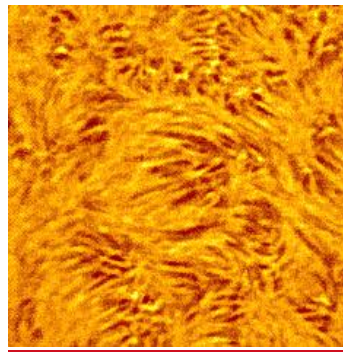
Credit: CESAR

- **Filaments:** a filament are prominences too, but viewed from a different perspective. A filament is against the Sun and appears darker while a prominence is viewed outside the solar disk. Some of them are not very long but others cover huge distances. As well as the prominences, they carry huge amount of material through the magnetic field lines.



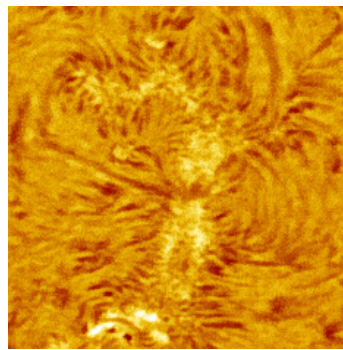
Credit: CESAR

- **Spicules:** a solar spicule is a jet of mass that moves upwards at a very high speed from the Sun's surface. There are thousands of them on the chromosphere so they are very easy to observe as elongated structures very close ones from each other.



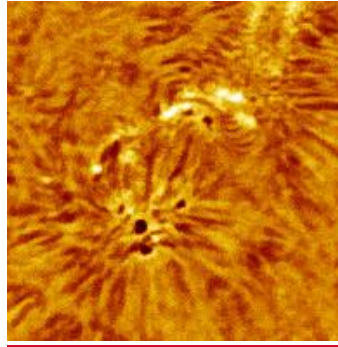
Credit: CESAR

- **Plages:** they are bright regions that appear near to the sunspots and they are usually bigger. They are typically associated with sunspot formation and also with concentrations of magnetic fields. They could also be observed without h-Alpha telescopes.



Credit: CESAR

- **Micro flares (or Ellerman bombs):** they are small flares that occur over the chromosphere. They appear as very brilliant dots on the surface. Through these flares, the Sun ejects very high quantities of energy.



Credit: CESAR

Material

1. CESAR Astronomical word list
2. CESAR Booklet
3. CESAR Formula sheet
4. CESAR images of the Sun
5. The software for this Science Case
6. Paper, pencil, ruler, calculator, protractor

Laboratory description and purpose

The main purpose of this laboratory is to calculate the differential rotation of the Sun by observing it and recognise the main features of the solar chromosphere. For this, you are going to take images of the Sun with the CESAR Solar Observatory. To be able to estimate the differential rotation you need a reference frame, which are the sunspots. You need to see their movement; otherwise it is impossible to do any calculations. By taking images over a time period it is possible to see how they move over the surface. From 3 to 7 days are enough to see any movement.

It is important to know if there are any sunspots on the Sun before attempting to take images. Visit SOHO's website and see the image of the Sun at the current date. If there are any sunspots present on the surface, then they probably will be there for the next days. The website for SOHO is <http://sohowww.nascom.nasa.gov/sunspots/>. Since the Sun is rotating around its own axis and its surface is constantly moving, the sunspots will obviously be at different locations each day.

Firstly you are going to do some measurements on the images to align them and once it has been done you are going to calculate the heliographic coordinates of at least two sunspots at different latitude to be able to appreciate the differential rotation. With this task you will see that is relative easy to learn about the structure of the Sun. Just attempting to the sunspots you will see that it is not a solid body and that it rotates at a very high rate considering its huge mass.

When you start with your calculations you can use whatever method you want, easy or a little more challenging one, it does not matter as long as you get a result that is reasonable and close to the real value.

We provide you the real value of the solar rate and also the rotation rate of all the rest of the planets so you could compare them attempting to their masses, volumes...

Finally using the h-Alpha images you should be able to recognize all the features on the chromosphere. It would be an easy task for you because you have an example for each feature in the “theoretical background” section. Take a look at all the images because some of these features may appear and disappear between each day. Furthermore, every feature may also appear with different sizes.

Laboratory execution

Start by reading the Booklet to understand the theory. By doing this, the laboratory will become much more easier to execute since you will have less questions and more understanding.

If your group has access to the Solar Observatory it is supposed that you have already choose a date for the beginning of the observation, and decided in how many days it will conclude. In the end, a start date and a finish date should be picked. An observation period of 3-7 days is a good example. It may seem to be many days but you will get a better visualization of all the Sun’s features.

Once you have the images of the Sun, it is time to start the measurements. You will need to write down the day and time the pictures you are using were taken and write down the position of the sunspots that appear on it. Follow your teacher instructions to first align the images and then calculate the heliographic coordinates of at least two sunspots at different latitudes as showed below. For example, two sunspots at 7 and 14 degrees is a good example; it doesn’t matter if the sunspots appear in different hemispheres, you just have to look at the absolute value of the latitude.

<u>Date/time</u>	<u>Sunspot</u>	<u>Longitude(°)</u>	<u>Latitude(°)</u>	<u>Observations</u>
<u>17/11/14 – 12:00</u>	<u>1</u>	<u>242,813</u>	<u>-18,209</u>	
<u>17/11/14 – 12:00</u>	<u>2</u>	<u>246,745</u>	<u>+12,699</u>	

Following the instructions included in the CESAR software for this Science Case you will be able to do all the measurements and you will also be able to align the images with a solar grid to easily determine the coordinates of any sunspot. Another way to do this is to plot the sunspots on a paper and use a protractor to project the Sun’s image on a board and then print the spots on a grid of the Sun, which your teacher is provided with if he consider this option.

When all the measurements have been done you have to calculate the angular velocity of the sunspots you choose. For this you just have to write down the longitudinal change in degrees for each pair of images and its corresponding time interval between each pair and all for each sunspot. Here is an example for the first one:

<u>Sunspot 1</u>		<u>Latitude 16</u>
<u>Pair</u>	<u>Longitude difference (°)</u>	<u>Time difference (days)</u>
<u>1-2</u>	<u>12,325</u>	<u>0,987</u>
<u>2-3</u>	<u>12,380</u>	<u>0,991</u>
<u>...</u>	<u>...</u>	<u>...</u>

Using the means of these values you should be able to calculate the angular velocity of each sunspot in units of degrees per day or degrees per second. As you know, you should appreciate the differential rotation of the Sun.

To compare these values with the angular velocity of other Solar System objects here is a table:

Planet's angular velocity	
Planet	Velocity (°/day)
Mercury	6,14
Venus	1,48
Earth	360
Mars	350,89
Jupiter	870,53
Saturn	818,96
Uranus	501,16
Neptune	536,27

Try to understand the whole concept behind this phenomenon and the calculations will be easy. Always ask your teacher for help.

Finally for the last task, you have to take the images that were taken by the h-Alpha telescope. Either if you use the CESAR software or not, you have to identify as much solar features as possible. If you use the software you will be asked for cataloguing each feature you found and a global summary will be automatically generated showing you all the prominences, filaments, and others that you recognized on the photos. Be careful and don't count the same features twice.

Teacher Guide for Kindergarten

Standard

K.LS3.1 Make observations to describe that young plants and animals resemble their parents.

Tennessee Academic Standards for Science: Page 19

Three-dimensional Learning Performance for Lesson

Students will ask descriptive questions after observing photographs of young organisms and their parents^{*} in order to show that young plants and animals resemble their parents^{**} highlighting the patterns of features used to determine if creatures are related based on similarities and differences.^{***}

Science and Engineering Practice for Lesson

*Asking Questions**

~~Students will be shown images of young animals and plants and then images of parent animals and plants, with the purpose of asking and refining scientific questions about the images. Initially, students will most likely ask broad questions, such as “Why do some animals look different as babies?” The teacher will then guide students to make more careful observations and notice particular features that could be evidence of a relationship between a young animal and their parent. The goal is to have students generate a more scientific line of questioning. For example, one student may be guided to ask, “Why does a baby chick have yellow feathers, but a grown-up chicken has brown/black/white feathers?” While another student may ask, “Why are their feathers different colors but their mouths/beaks look similar?” By refining their questions to focus more on similar and different features, students make more refined observations about which features provide evidence that young plants and animals resemble their parents.~~

Disciplinary Core Idea for Lesson

*Life Science 3: Heredity***

~~“Organisms have characteristics that can be similar or different. Young animals are very much, but not exactly, like their parents and also resemble other animals of the same kind. Plants also are very much, but not exactly, like their parents and resemble other plants of the same kind.”~~

A Framework for K–12 Science Education: Page 158

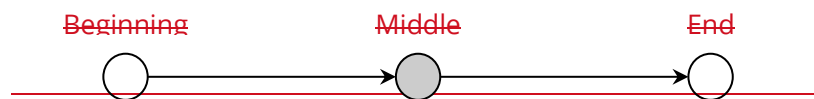
Crosscutting Concept for Lesson

Patterns ***

Students observe that plants and animals of a certain species display a pattern of similar features or characteristics. These patterns are typically even more similar when observing organisms that are part of the same family.

Prior Knowledge

Location Within Instructional Unit



- Concepts taught prior to this lesson:
 - We use our senses to notice things, such as features on a plant or animal
 - Young animals often resemble their parents
 - We determine this by looking for how similar their features are to each other

Materials

- Tree Detective Sheet
- Animal Matching Cards
- "Animal Parents and Kids" YouTube slideshow
- Image of Mature Pine Tree
- Image of Pine Tree Sapling
- Image of Detective (Cartoon)
- Image of Kid Detective
- Optional "Detective" Lens (hand lens) for students

Lesson Sequence Outline

1. Review that young animals often resemble their parents
2. Complete matching animal activity
3. Guide students to ask a similar question about plants (e.g., "Do young plants resemble their plant parents?" or "Do all living things resemble their parents?")
4. Discuss how a detective is a special kind of policeman that investigates things to get all the possible clues, which they then use to solve a case or answer a question

5. ~~Set up students for activity by having them get their 'detective lens' out so they can think like a detective~~
6. ~~Have students engage with the *Tree Detective Sheet* and collect for formative assessment~~

Instructional Notes

The teacher begins the lesson by reviewing from a previous lesson with students that young animals often resemble their parents. To do this, the teacher (re)plays the YouTube slideshow "[Animal Parents and Kids](#)," and pauses the video at each slide so that they can call on students to identify similar features that provide evidence that these animals are related. For example, students may say "The parent porcupine and baby porcupine both have spiky backs!" or "The puppy has the same color brown fur as its parent." A focus for this part of the lesson is to have students generate evidence that describes how the animals are similar, rather than just declaring "They look the same!"

Once the slideshow is over, the teacher will have students demonstrate their understanding of animals and their parents by having them match the parent animal with its kid using the *Animal Matching Cards*. The cards should be copied, cut out ahead of time, and laminated if possible. Students can work in pairs to match their cards. Whenever a student matches animals together, they should describe evidence for why they think those animals are related. For example, the student may say, "I think the cat and the kitten are related because they have the same grey colored fur." Again, the focus for this activity is to have students work on identifying specific features that are similar between animals.

After students have completed the *Animal Matching Cards* activity, the teacher gathers students back together and tells them a story. In the story, the teacher was out walking in nature when they noticed a big pine tree. The teacher displays the [Image of Mature Pine Tree](#) to students so that they can understand what the tree looks like. Then, the teacher continues saying that during that same walk, they noticed another tree that looks similar in some ways, but different in other ways. The teacher displays the [Image of Pine Tree Sapling](#) to students and asks students if they notice what is similar about these two pictures. As students offer suggestions, such as, "They both have pine needles!" the teacher pretends to do some deep thinking before saying to students, "You guys are noticing some really interesting similarities between these two trees. Hearing all your thoughts is causing me to have some questions about these trees now. Does anyone else have a question or is maybe wondering something about these two trees?" From here, the teacher guides students to

ask questions about the trees with the goal of having students wonder if the trees could possibly be related.*

**Note: The goal of having students ask questions about the trees is to hopefully have a student ask if the two trees are possibly related, like the animals in the matching game and on the slideshow. Students may require some scaffolding to arrive at this question, so the teacher can guide the class to ask these sorts of questions by suggesting things like “I notice several of you are asking questions about how one tree is little and one tree is really big. When we looked at animals and their parents on the slideshow earlier, was one of them always big and one of them always little? (Yes!) What could this mean about the big tree and the little tree then? Are you saying that they could be related?”*

Once students are primed to examine if the trees are possibly related to each other, the teacher tells students, “Trees can be harder to figure out than animals. This is because trees in some ways are very different than animals, but if we go searching for clues very carefully, we might find some evidence that helps us find out the real answer.” The teacher then displays the Image of Detective (Cartoon) and asks students if they know what a detective does. If they do not know, the teacher explains that a detective is someone who looks very carefully for clues to help them solve something. The teacher explains that a detective is usually shown with a hand lens because they use tools to help them look for clues. The teacher emphasizes that most importantly, though, a detective uses those clues to ask “Detective Questions.” These are questions related to the mystery that the detective is trying to solve. The teacher can give an example of this to students, such as “For example, when I was walking on the nature trail the other day and saw those two trees I thought to myself ‘I noticed those two trees look like a parent tree and a baby tree because they have the same type of needle leaves, but one was really big and the other one was really small. I wonder if these trees are related like the animals we saw earlier?’”

The teacher can model this further by working with the class to point out any other similarities that they notice that might be evidence that two trees are related. Some similarities include: The leaves appear the same, they both have leaves growing from a branch, they both are growing up out of the ground, they both have green leaves and a brown trunk, etc. This may need to go on for a few minutes to give students a chance to notice and describe similarities between the two trees.

When students are ready to move on, the teacher informs them that for this next activity, the students will need to become detectives to solve the mystery of whether the following trees could be related. In order to do this, they must look for evidence (similar features) in the pictures on the activity sheet they will receive. This means they will need to use their

“Detective Hand-lens” to make careful observations of some images of trees and determine how the trees are similar. The teacher can display the Image of Kid Detective to help solidify this point. In order to work as “tree detectives,” the students will each receive a copy of the Tree Detective Sheet as well as an optional hand-lens. Students will need to use the sheet to do the following:

1.—Page 1

A. Look closely at two images of trees to notice how they are similar and how they are different

i.—Students can use their ‘detective hand-lens’ to help them focus on the images and hone in on specific features of the trees

B. Describe one or two ways that the trees are similar

C. Describe one or two ways that the trees are different

2.—Page 2

A. Use the images of two “mystery” plants to notice how they are similar and how they are different

B. Formulate their own “detective question” about the two plants

i.—Examples:

1.—Are the plants related?

2.—Will the smaller/“baby” plant grow up to be like the bigger/“parent” plant?

3.—Will the “arms”/branches on the smaller plant grow?

4.—Do the plants have the same parts?

As students are working, the teacher is monitoring students and assisting as needed. Once finished, the teacher can collect the sheets to look over student responses as a formative assessment of student understanding.

Student Responses and Formative Assessment

Young Birch/Mature Birch observations:

- Similarities include: trunks are both white-ish, both types of trees grow out of the ground, both trees have leaves, etc.
- Differences include: Young birch trunks are more skinny than mature birch trunks, mature birch has more leaves than young birch, mature birch trunks form a ‘Y’ shape and young birch trunks go up in a straight line, etc.

Mystery Desert Plants

The images are of a young Saguaro cactus and a mature Saguaro cactus.

Students should notice how similar the two plants appear in the pictures. The branches on the young Saguaro will develop over time to look more like the mature Saguaro.

As students are deciding on a 'detective' question related to these plants, they should choose the question that asks if the plants come from the same family based on their similarities.

Citations and Resources

Image of Pine Tree:

<https://www.flickr.com/photos/forestfortrees/4359236961/>

Image of Pine Tree Sapling:

<https://www.flickr.com/photos/96713863@N00/1290126678/>

Image of Detective (Cartoon):

<https://pixabay.com/en/inspector-man-detective-male-160143/>

Image of Kid Detective:

<https://pixabay.com/en/detective-magnifying-glass-viewing-788592/>

Animal Matching Cards

Image of Goat:

<https://pixabay.com/en/goat-billy-goat-cute-goat-2644998/>

Image of Kid:

https://commons.wikimedia.org/wiki/File:Hausziege_04.jpg

Image of Deer:

<https://pixnio.com/fauna-animals/deers/deer-female-in-wild>

Image of Fawn:

<https://www.flickr.com/photos/slopjop/828388505>

Image of Kangaroo:

https://en.wikipedia.org/wiki/File:Kangaroo-in-flight_cropped.jpg

Image of Joey:

https://commons.wikimedia.org/wiki/File:Macropus_rufogriseus_rufogriseus_Joey_1.jpg

Image of Cat:

<https://pxhere.com/en/photo/986828>

Image of Kitten:

<https://pixabay.com/en/cat-tomcat-kitty-kitten-2866167/>

Image of Duck:

https://commons.wikimedia.org/wiki/File:White_domesticated_duck_stretching.jpg

Image of Duckling:

<https://pxhere.com/en/photo/994760>

Image of Dog:

<https://pixabay.com/en/dog-labrador-light-brown-pet-1210559/>

Image of Puppy:

<https://commons.wikimedia.org/wiki/File:Too-cute-doggone-it-video-playlist.jpg>

Tree Detective Sheet

Image of Mature Birch Tree retrieved from:

<https://pixabay.com/en/nature-outdoors-woods-birch-tree-3070475/>

Image of Young Birch Tree retrieved from:

https://commons.wikimedia.org/wiki/File:Birch_forest_with_birds_nest_Gullmarsskogen.jpg

Image of Walnut Tree retrieved from:

<https://www.flickr.com/photos/brighton/4621671958>

Image of Young Saguaro Cactus retrieved from:

<https://www.nps.gov/tont/learn/nature/saguaro.htm>

Image of Mature Saguaro Cactus retrieved from:
<https://www.nps.gov/orpi/learn/nature/saguaro-cactus.htm>